



## 1.6

## Polynomial Functions and End Behavior

### Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1

| ECHS Mathematics

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#### Student Information

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

#### Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

#### Readiness

1. Write one precise sentence explaining this principle: Even degree means both ends point in the same vertical direction.  
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2. Write one precise sentence explaining this principle: Odd degree means the ends point in opposite vertical directions.  
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3. Write one precise sentence explaining this principle: The sign of the leading coefficient controls the right end.  
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4. Write one precise sentence explaining this principle: End behavior does not determine the number or locations of turning points.  
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**Core**

5. Describe the end behavior of  $P(x) = -4x^7 + 2x^3 - 1$ .
6. Can a positive-leading even-degree polynomial have a local maximum?
7. Compare  $3x^6 - 10x^5$  and  $3x^6 + 1000x$  for large  $|x|$ .
8. State what the formula  $P(x) = a_nx^n + \dots$  tells you and name any required restriction.
9. State what the formula  $P(x) \sim a_nx^n$  as  $|x| \rightarrow \infty$  tells you and name any required restriction.
10. Create a second representation (table, graph, formula, or context) for one central relationship in Polynomial Functions and End Behavior. State its domain and justify one key feature.

**Medium**

11. Describe both ends of  $4x^8 - 3x + 1$  in words and limit notation.
12. Describe both ends of  $-2x^5 + x^4$  in words and limit notation.
13. Describe both ends of  $-x^{10} + 100x^9$  in words and limit notation.
14. Describe both ends of  $7 - 3x^3$  in words and limit notation.

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15. Describe both ends of  $0.2x^6 - 10x^2$  in words and limit notation.

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**Hard**

16. Two polynomials have the same degree parity and leading coefficient sign. What end behavior must they share, and what may differ?

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17. Find all signs/parities consistent with  $P(x) \rightarrow -\infty$  as  $x \rightarrow -\infty$  and  $P(x) \rightarrow \infty$  as  $x \rightarrow \infty$ .

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18. Construct a degree-6 polynomial with both ends down and exactly two real zeros.

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19. Which term controls end behavior of  $(3 - x)^7 + 5x^6$ ?

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**Difficult**

20. A polynomial satisfies  $P(x)/x^9 \rightarrow -4$  as  $x \rightarrow \infty$ . What can be concluded?

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21. Can a polynomial with both ends up have an odd number of distinct real zeros, all simple? Explain.

22. Construct two polynomials with identical end behavior and real zeros but different multiplicities.

### Challenge / HOT

23. Prove that a polynomial with odd degree and real coefficients must have opposite end behavior.

24. Can two polynomials differ by an unbounded amount but have the same end behavior? Give an example.

### AP-Style Multiple Choice

25. End behavior of  $-3x^6 + 2x$  is  
(A) both up

- (B) both down
  - (C) left down/right up
  - (D) left up/right down
26. Right end rises and left end falls means
- (A) even positive
  - (B) even negative
  - (C) odd positive
  - (D) odd negative
27. Which feature does not affect polynomial end behavior?
- (A) degree parity
  - (B) leading coefficient sign
  - (C) constant term
  - (D) leading term
28. As  $|x| \rightarrow \infty$ ,  $2x^5 - 7x^4$  behaves like
- (A)  $-7x^4$
  - (B)  $2x^5$
  - (C)  $2x$
  - (D)  $x^9$
29. A polynomial with both ends down must have
- (A) odd degree
  - (B) negative leading coefficient and even degree
  - (C) positive leading coefficient
  - (D) exactly two zeros
30. Which limit is correct for  $x^8 - 10x^7$ ?
- (A)  $x \rightarrow -\infty, P \rightarrow -\infty$
  - (B)  $x \rightarrow \infty, P \rightarrow -\infty$
  - (C) both ends  $+\infty$
  - (D) both ends 0

### AP-Style Free Response

31. A polynomial graph has left end up and right end down, zeros at  $-2$  (touches) and  $1$  (crosses). (a) State degree parity and leading sign. (b) Give minimum multiplicities. (c) Construct a least-degree example.

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32. Let  $P(x) = a(x - r_1)^{m_1} \cdots (x - r_k)^{m_k}$ . Explain how degree,  $a$ , and multiplicities jointly determine end behavior and sign changes.

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Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

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Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.